

Autonomous Driving Sensor Data Report (Dataset 3)

Detailed Breakdown of matlab_data_3 Export

1. Overview & Data Source Variations

This report details the structural analysis of the 'matlab_data_3' directory. Unlike previous datasets, this simulation run was exported with a distinct variation: Camera data was completely omitted. This forces the downstream algorithms to rely purely on Radar and LiDAR for object detection and trajectory prediction.

2. Radar Data (r_data.mat)

Format: MATLAB Struct (.mat)

Contents: 50 independent radar detections.

Fields Extracted:

- Time: Simulation timestamp of detection.
- SensorIndex: Unique ID of the radar sensor.
- ObjectClassID: Numerical classification of the detected object.
- Measurement: Numerical array typically containing [Azimuth, Range, Range_Rate].

Interpretation: Since camera data is missing, the Radar acts as the primary source for calculating relative velocity (Range_Rate) to compute Time-to-Collision (TTC).

3. Camera Data (MISSING)

Status: OMitted / Not Exported.

The c_data.mat file is completely absent from this dataset. Autonomous Emergency Braking (AEB) systems handling this dataset must implement Sensor Fusion algorithms that combine Radar (for speed) and LiDAR (for volumetric size/bounding boxes) without optical classification confirmation.

4. LiDAR Data (Lidar_PCD_Data/)

Format: Standard Point Cloud Data (.pcd) Files

Contents: 1001 individual frames of 3D spatial points.

Data Volume: The presence of 1001 frames (compared to 800 in previous sets) indicates either a longer simulation scenario or a higher frequency capture rate.

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Processing Pipeline: Python's 'open3d' library should be used to load the point clouds, apply RANSAC to remove the ground plane, and DBSCAN to cluster points into 3D Bounding Boxes to compensate for the missing optical camera detections.

5. Conclusion & Actionable Steps

The proprietary MATLAB files have been unpacked. The radar detections have been extracted into Results/radar_detections.csv. The 1001 LiDAR .pcd files are ready for immediate Python processing. Algorithms built for this dataset must account for the lack of camera telemetry.